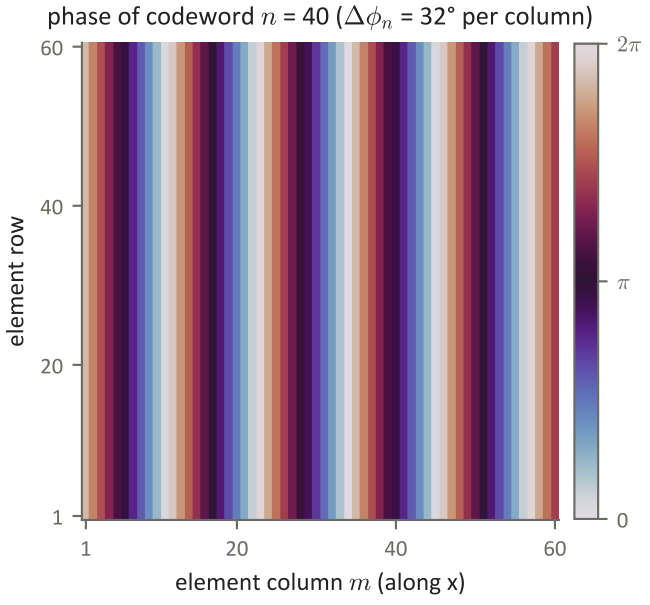
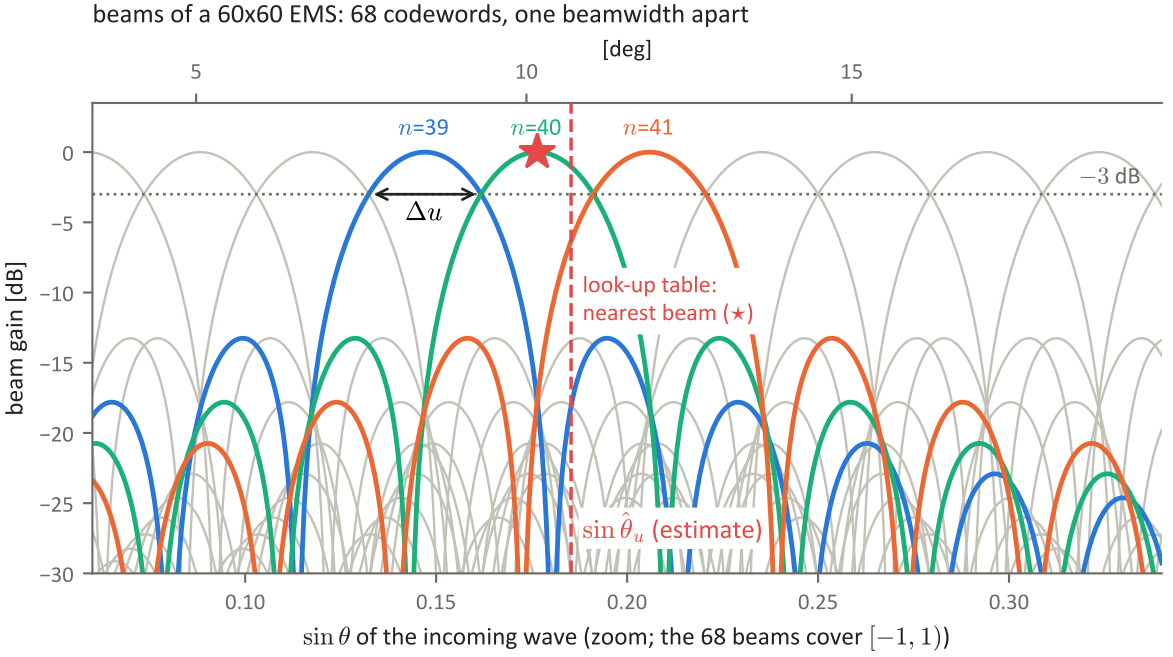

phase of codeword n = 40 ($\Delta\phi_n$ = 32° per column)
element row
element column m (along x)
60
40
20
1
1
20
40
60
2π
π
0


beams of a 60x60 EMS: 68 codewords, one beamwidth apart
[deg]
5
10
15
n=39
n=40
n=41
-3 dB
Δu
look-up table:
nearest beam (⋆)
$\sin\hat{\theta}_u$ (estimate)
beam gain [dB]
0
−5
−10
−15
−20
−25
−30
0.10
0.15
0.20
0.25
0.30
$\sin\theta$ of the incoming wave (zoom; the 68 beams cover $[-1, 1)$)